

Commutativity of the natural one-dimensional quaternion convolution

A complete answer to the open problem in arXiv:2402.06645

Autonomous solve-first investigation, lane 12

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Abstract

Gupta and Verma ask for a nontrivial necessary-and-sufficient condition for commutativity of the natural quaternion convolution of Akila and Roopkumar. We give a complete classification. Two quaternion-valued functions commute under this convolution exactly when the imaginary parts of their quaternion Fourier transforms are collinear at every frequency. Equivalently, the two Fourier values lie pointwise in a common complex slice of the quaternion algebra. We prove the equivalence from the convolution theorem and Fourier injectivity, derive three explicit real component identities and two time-domain convolution identities, and exhibit broad families beyond the sufficient cases recorded in the survey.

1 The question and the answer

Fix quaternion units i, j, k with $ij = k$ and write every quaternion-valued function uniquely as

$$f = f_1 + jf_2, \quad f_1, f_2 : \mathbb{R} \rightarrow \mathbb{C}_i,$$

where $\mathbb{C}_i = \mathbb{R} + i\mathbb{R}$. For a complex function h , put $\check{h}(x) = h(-x)$. The natural convolution from Akila–Roopkumar [2], in the notation of the source survey, is

$$\begin{aligned} (f \circledast g)(x) &= (f_1 * g_1)(x) - (\check{\bar{f}}_2 * g_2)(x) \\ &\quad + j((\check{\bar{f}}_1 * g_2)(x) + (f_2 * g_1)(x)). \end{aligned} \tag{1}$$

The quaternion Fourier transform is componentwise,

$$\widehat{f}(\xi) = \widehat{f}_1(\xi) + j\widehat{f}_2(\xi), \tag{2}$$

and its decisive compatibility with (1) is

$$\widehat{f \circledast g}(\xi) = \widehat{f}(\xi)\widehat{g}(\xi). \tag{3}$$

The survey notes that $\circledast$ is generally noncommutative and asks for a nontrivial necessary-and-sufficient condition for $f \circledast g = g \circledast f$ [1, p. 26]. Its final sentence prints an ordinary star, but the immediately preceding definition, convolution theorem, and known sufficient cases unambiguously identify (1) as the intended operation. Figure 1 reproduces the source passage.

The theorem is stated on $L^1_{\mathbb{H}}(\mathbb{R})$, where every term in (1) lies in L^1 and the Fourier transform is pointwise defined and injective. It therefore also applies on the common $L^1 \cap L^2$ domain in the source's L^2 formulation.

Theorem 1 (complete commutativity criterion). *Let $f, g \in L^1_{\mathbb{H}}(\mathbb{R})$. The following are equivalent.*

- (i) $f \circledast g = g \circledast f$ almost everywhere.
- (ii) $\widehat{f}(\xi)\widehat{g}(\xi) = \widehat{g}(\xi)\widehat{f}(\xi)$ for every $\xi \in \mathbb{R}$.
- (iii)

$$\operatorname{Im} \widehat{f}(\xi) \times \operatorname{Im} \widehat{g}(\xi) = 0 \quad (\xi \in \mathbb{R}). \tag{4}$$

where pure imaginary quaternions are identified with $\mathbb{R}^3$ in the basis (i, j, k) .

Also if we take $f = f_1 + jf_2$ and $g = g_1 + jg_2$, then we have

$$\begin{aligned} \overline{(f \star g)}(\xi) &= \left[\hat{f}_1(\xi) \hat{g}_1(\xi) - \hat{f}_2(\xi) \hat{g}_2(\xi) \right] + j \left[\hat{f}_2(\xi) \hat{g}_1(\xi) - \hat{f}_1(\xi) \hat{g}_2(\xi) \right] \\ \hat{f}(\xi) \hat{g}(\xi) &= \left[\hat{f}_1(\xi) \hat{g}_1(\xi) - \hat{f}_2(\xi) \hat{g}_2(\xi) \right] + j \left[\hat{f}_2(\xi) \hat{g}_1(\xi) - \hat{f}_1(\xi) \hat{g}_2(\xi) \right]. \end{aligned}$$

Thus $\overline{(f \star g)}(\xi) = \hat{f}(\xi) \hat{g}(\xi)$ if $\bar{\hat{f}}_1(\xi) = \hat{f}_1(\xi)$ and $\bar{\hat{f}}_2(\xi) = \hat{f}_2(\xi)$. Bahri et al. [33] also studied one-dimensional quaternion FT with some convolution $\star$ for which $f \star g \neq g \star f$ in general and also faces the same problem as that in [93]. Akila et al. [5] introduced a novel natural convolution given as

$$(f \otimes g)(x) = (f_1 \star g_1)(x) - (\bar{\hat{f}}_2 \star g_2)(x) + j \left((\bar{\hat{f}}_1 \star g_2)(x) + (f_2 \star g_1)(x) \right) \quad (142)$$

and proved that $f \otimes \delta \in L_{\mathbb{H}}^p(\mathbb{R})$, $\alpha(f \otimes g) = (\alpha f) \otimes g$, $(f + g) \otimes \delta = f \otimes \delta + g \otimes \delta$ and $(f \otimes g) \otimes \epsilon = f \otimes (\delta \otimes \epsilon)$ for all $\alpha \in \mathbb{H}$, $\delta, \epsilon \in L_{\mathbb{H}}^1(\mathbb{R})$, and $f \in L_{\mathbb{H}}^p(\mathbb{R})$, $p = 1, 2$. They also proved the convolution theorem $\mathcal{F}(f \otimes g) = \mathcal{F}(f) \mathcal{F}(\delta)$, $\delta \in L_{\mathbb{H}}^1(\mathbb{R})$ and $f \in L_{\mathbb{H}}^2(\mathbb{R})$, and hence $\otimes$ defined in (142) serves as a better convolution structure for the transform defined in (140) than the one given in (141) and also in [33]. In general, $\otimes$ is non-commutative, i.e., $f \otimes g$ need not be equal to $g \otimes f$. If both f and g are complex-valued or if one of them is a real-valued even function, then we get $f \otimes g = g \otimes f$. Finding a suitable non-trivial necessary and sufficient condition on f and g to get $f \star g = g \star f$ is still an interesting open problem. The definition (140) together with the convolution structure (142) and its properties for the arbitrary dimension d , i.e., for the functions defined on $\mathbb{R}^d$ can be found in [6].

7 Quaternion Fractional Fourier Transform (QFrFT)

Figure 1: Definition (142), its Fourier product theorem, and the open commutativity problem in arXiv:2402.06645, p. 26.

- (iv) At each frequency ξ , the two values $\hat{f}(\xi)$ and $\hat{g}(\xi)$ belong to a common complex slice $\mathbb{C}_{u(\xi)} = \mathbb{R} + \mathbb{R}u(\xi)$ for some pure unit quaternion $u(\xi)$, with arbitrary $u(\xi)$ allowed when both values are real.

More explicitly, write

$$\hat{f} = a + jb, \quad \hat{g} = c + jd,$$

with $a = a_0 + ia_1$, $b = b_0 + ib_1$, $c = c_0 + ic_1$, and $d = d_0 + id_1$. Then the equivalent pointwise scalar criterion is

$$\boxed{b_1 d_0 - b_0 d_1 = 0, \quad a_1 d_1 - b_1 c_1 = 0, \quad a_1 d_0 - b_0 c_1 = 0.} \quad (5)$$

All eight real functions in (5) are evaluated at the same frequency ξ , and the equations must hold for every ξ .

2 Proof

We isolate the elementary quaternion fact that makes the criterion exact.

Proposition 2 (centralizers in $\mathbb{H}$). For $p = p_0 + p_1 i + p_2 j + p_3 k$ and $q = q_0 + q_1 i + q_2 j + q_3 k$, put $\mathbf{p} = (p_1, p_2, p_3)$ and $\mathbf{q} = (q_1, q_2, q_3)$. Then

$$pq - qp = 2 \mathbf{p} \times \mathbf{q}. \quad (6)$$

Consequently, p and q commute if and only if their imaginary vectors are collinear. Equivalently, they lie in one complex slice $\mathbb{R} + \mathbb{R}u$.

Proof. The standard scalar–vector multiplication rule is

$$(p_0 + \mathbf{p})(q_0 + \mathbf{q}) = (p_0 q_0 - \mathbf{p} \cdot \mathbf{q}) + (p_0 \mathbf{q} + q_0 \mathbf{p} + \mathbf{p} \times \mathbf{q}).$$

Interchanging p and q preserves every term except the cross product, which changes sign, proving (6). A vanishing cross product is precisely collinearity. If a nonzero imaginary vector occurs, normalize its common direction to obtain u ; if both vanish, every u works. $\square$

Proof of Theorem 1. Apply the convolution theorem (3) to both orders:

$$\widehat{f \circledast g} = \widehat{f} \widehat{g}, \quad \widehat{g \circledast f} = \widehat{g} \widehat{f}. \quad (7)$$

If the convolutions agree, their Fourier transforms agree, so (i) implies (ii). Conversely, (ii) and (3) give equal quaternion Fourier transforms. Componentwise injectivity of the complex Fourier transform on $L^1(\mathbb{R})$ yields equality of the two quaternion-valued convolutions almost everywhere. Thus (i) and (ii) are equivalent.

At each fixed ξ , Proposition 2 identifies (ii) with (iii), and also identifies (iii) with (iv). It remains only to expand (iii). Because

$$a + jb = a_0 + a_1 i + b_0 j - b_1 k,$$

the two imaginary vectors are

$$(a_1, b_0, -b_1), \quad (c_1, d_0, -d_1).$$

Their cross product is

$$(b_1 d_0 - b_0 d_1, a_1 d_1 - b_1 c_1, a_1 d_0 - b_0 c_1),$$

so its vanishing is exactly (5). $\square$

There is a useful second algebra check. Since $jz = (\bar{z})j$ for $z \in \mathbb{C}_i$,

$$(a + jb)(c + jd) = ac - (\bar{b})d + j((\bar{a})d + bc). \quad (8)$$

Subtracting the reverse product in (8) gives the equivalent pair of complex equations

$$(\bar{d})b - (\bar{b})d = 0, \quad (\bar{a})d - da + bc - (\bar{c})b = 0. \quad (9)$$

This independently reproduces all three real equations in (5).

3 Time-domain form and genuinely new examples

Direct subtraction in (1), using commutativity of ordinary complex convolution, gives an exact time-domain version.

Corollary 3 (time-domain criterion). *Under the hypotheses of Theorem 1, $f \circledast g = g \circledast f$ if and only if*

$$\check{\check{f}}_2 * g_2 = \check{\check{g}}_2 * f_2, \quad (10)$$

$$\check{\check{f}}_1 * g_2 + f_2 * g_1 = \check{\check{g}}_1 * f_2 + g_2 * f_1. \quad (11)$$

The frequency criterion is more structural: it says that the slice may vary with ξ , and it automatically handles frequencies at which one transform is real or zero. In particular, it strictly exceeds the two sufficient cases stated in the survey.

Corollary 4 (a nontrivial commuting family). *Fix a pure unit quaternion u , and let $\phi, \psi \in L^1(\mathbb{R})$ be real even functions. For arbitrary real constants r, s, t, v , the functions*

$$f = (r + su)\phi, \quad g = (t + vu)\psi. \quad (12)$$

commute under $\circledast$.

Proof. The Fourier transforms $\widehat{\phi}$ and $\widehat{\psi}$ are real-valued. Hence the Fourier values of both functions lie in the fixed commutative slice $\mathbb{C}_u$, and Theorem 1 applies. $\square$

For example, take

$$u = \frac{i+j}{\sqrt{2}}, \quad f = (1+u)e^{-x^2}, \quad g = (2+3u)e^{-2x^2}. \quad (13)$$

Neither function is $\mathbb{C}_i$ -valued and neither is real-valued, yet $f \circledast g = g \circledast f$. Thus this is not covered by either easy case in the survey. More generally, Theorem 1 allows the common slice $\mathbb{C}_{u(\xi)}$ to rotate with frequency, subject only to the Fourier transforms coming from the chosen function class.

Conclusion. The open problem has an exact answer: natural-convolution commutativity is neither an ad hoc parity condition nor global membership in one fixed copy of $\mathbb{C}$. It is precisely pointwise collinearity of the two imaginary Fourier vectors. Conditions (5), (9), and (10)–(11) are respectively real-frequency, complex-frequency, and time-domain versions of the same classification.

References

- [1] B. Gupta and A. K. Verma, *A mathematical survey on Fourier type integral transform and their offshoots: windowed Fourier transform, wavelet transform and Stockwell transform*, arXiv:2402.06645 (2024).
- [2] L. Akila and R. Roopkumar, *A natural convolution of quaternion valued functions and its applications*, Applied Mathematics and Computation 242 (2014), 633–642.